SubwayHorse's Design Doc for P04: Makers Makin' It, Act II -- The Seequel

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PROJECT NAME: CityCash
TARGET SHIP DATE: 2026-04-22

Summary

Our project aims to show the job prospects of different NYC Government employees. The user will be presented with a color-coded map of the city. The user will be able to compare salaries based on location. They can also look at the evolution of salaries over a specific time period. They can also view data like salary vs department or job title by generating their own graphs. Our goal is to make it easy to compare job salaries, and judge job prospects based on certain factors.

Program Components

Frontend:

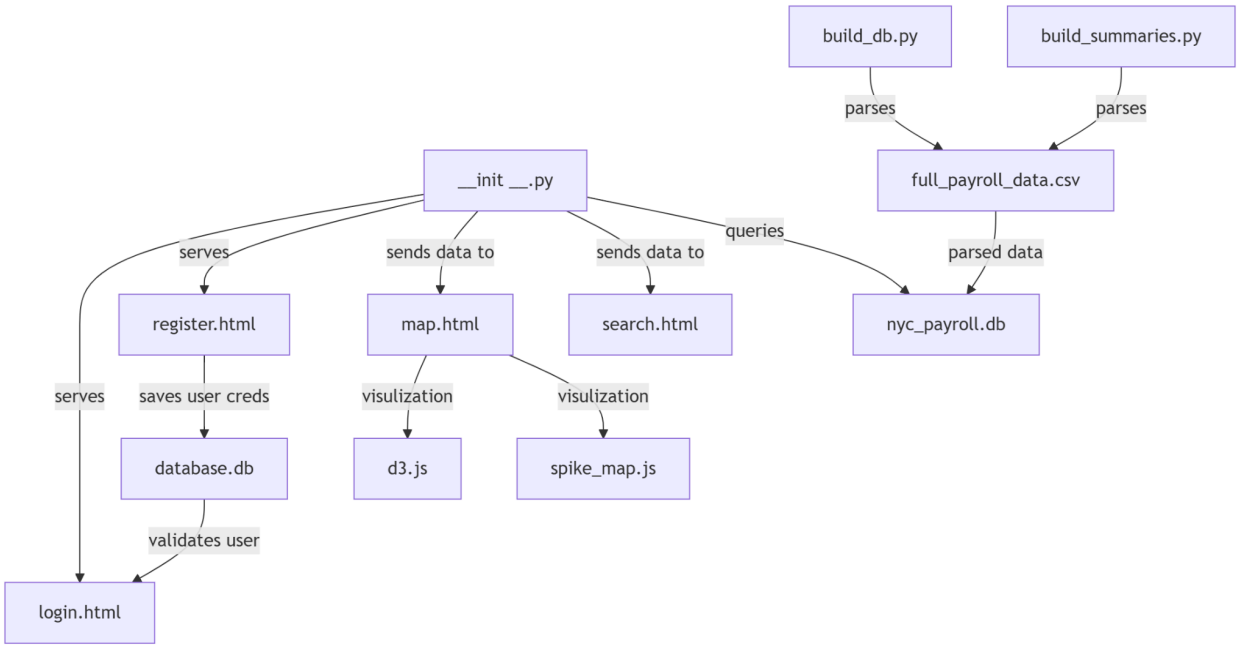
- HTML (frontend display)
  - login.html
  - register.html
  - map.html: Contains a map of NYC. There will also be a dropdown menu where you can find data you want (like job X's avg. salaries) and plot it against other data. Users can dynamically create graphs, or can compare stats across boroughs.
  - search.html: a filter function to dynamically display individual data points in a table
- CSS
  - Tailwind → style **login.html**, **register.html**, **search.html**, and **map.html**
- Javascript (data visualization)
  - d3.js → responsible for the custom chart in **map.html**
  - spike_map.js → responsible for the spike map in **map.html**

Backend:

- sqlite3 (data storage system)
  - payroll_data → accessed by **map.html** and **search.html** to display data

    - user_data → accessed by **register.html** and **login.html** to facilitate user accounts
- Python (application layer)
    - __init__.py
    - build_db.py
    - build_summaries.py
- Flask (web server/delivery framework)

Component Map



Database Organization

We are using SQLite3, a relational database, for USER_DATA, because that's the way it's been done with proven success.

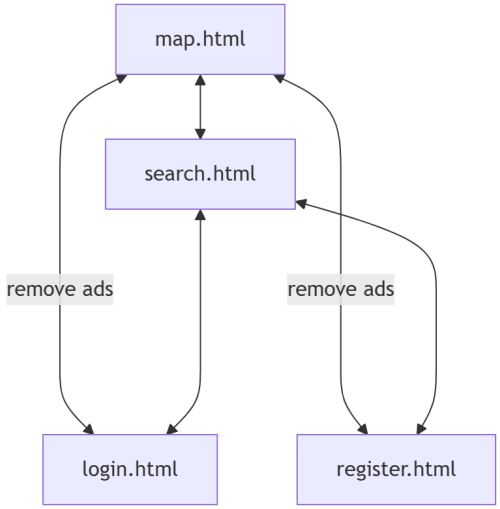
We are also using SQLite3 for PAYROLL_DATA in order to query the database and send to the frontend only the data that is needed, instead of the entire 1.1 GB CSV file.

We will keep all the data from the dataset, because we believe that everything here is of use to someone (examples include looking for leave status of Mr. Moran (well not anymore), crazy OT hours, corruption or shenanigans in total other pay, etc, etc)

PAYROLL_DATA	
fiscal_year	int
payroll_number	int
agency_name	string
last_name	string
first_name	string
middle_init	string
agency_start_date	date
work_location	string
title_description	string
leave_status	string
base_salary	int
pay_basis	string
regular_gross_paid	int
ot_hours	int
total_ot_paid	int
total_other_pay	int

USER_DATA		
username	string	PK
password	string	

Site Map



Task Breakdown

steven	PM, javascript/data visualization
alex	payroll_data setup, javascript
cindy	user_data setup/accounts functionality, css/tailwind, javascript/data visualization
jeff	javascript/data visualization

Front-end Framework

Tailwind: because of our familiarity with tailwind, it is the most efficient for our purposes, some of which include:

- Navbar → for easy navigation through the site
- Cards → for register and login styling
- Grids → for organization

APIs

None

Data Visualization Library

PROS: d3.js	**CONS: d3.js**
<u>More versatility</u> d3 enables custom shapes (which we will need for the map), and chart.js does not.	<u>Higher effort</u> chart.js offers a good selection of built-in chart types, but selecting d3 for custom chart types means putting in more effort.
<u>More engaging visuals</u> d3 also has higher potential for interactive elements, which our project emphasizes.	<u>Steep Learning Curve</u> d3 is definitely more complicated to wield well than chart.js.

Because the highlight (and purpose) of our project is the data visualizations, **our team elected to use d3.js** because it supports our vision. On its website are several examples that we might want to implement in our map of NYC. For instance, they have a map of the US with regional coloring based on obesity, as well as another one where you can zoom into a US state when clicked on.